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MES – BUSINESS CASE

GATES INDIA LALRU

SMART CONTROLS
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AGENDA

1. Plant Study Report	Machines understanding
	Process understanding
2. Proposed Solutions	Op Centre MES Solution
	Smart MES Solution

NO OF MACHINES OR STATIONS

Serial No	Area	No of Machines	Description
1	RMS Booking	1	Raw material storage, receipt, and placement in the storage area.
2	RMS Issue	2	Issue material to shop floor from raw material storage.
3	Carbon Silica & Other	12	12 feeders feed material.
4	Oil	10	Oil stations for specific materials.
5	Auto Chemical	2	Approx. 50 bins for feeding in each small chemicals.
6	Bale Cutter	2	Cutting polymer into separate lots.
7	Mixer Feeding	2	Operator feeds polymer/compound based on recipe.
8	Mixer Batchoff	2	Load batch output onto skid.
Total	8	33	

RMS REQUIREMENT

ONLY FOR MIXING AND SMALL CHEMICALS

UNDERSTANDING

- GRN will be booked in ERP System.
- ERP system will send booked RM Materials related to mixing process to MES.
- In Shop floor storage area will be identified and marked for easy storage and pickup.
- In case GRN is cancelled or Quality is changing ERP will share the information with MES.

SOLUTION KEY FEATURES

- MES will allow user to split material quantity into sub lots for better traceability.
- MES will allow user to print QR code.
- MES will allow user to select the storage area.
- MES will maintain RMS inventory material wise location wise and aging wise.

REQUIRED HARDWARE

- 1 PC for MES client, Where user can open the MES for access GRN data and give print command.
- 2 Printer one is for sticker and another one is for tag.
- LAN Network is required to connect these PC and Printers with MES server.

RMS REQUIREMENT

ONLY FOR MIXING AND SMALL CHEMICALS

UNDERSTANDING

- User will create RM Issue list based on production requirement.
- On Issue list items are transferred to shopfloor MES will Update the inventory and sync with ERP.

SOLUTION KEY FEATURES

- MES RMS module will have option to create issue list.
- Required Issue list will be displayed on mobile scanner for end user to pick up the materials.
- On material scanned , System will Validate material, quantity, aging, and FIFO.
- Based on issue slip, user issues the material with MES mobile scanner.
- After all required materials are Issued, MES will have option to complete the Issue and sync the inventory.

REQUIRED HARDWARE

- 2 Wi-Fi Mobile scanner required.
- Issue slip user can generate in RMS PC.
- Wi-Fi Network is required in RMS Store.

CARBON SILICA & OTHER FEEDERS REQUIREMENT

UNDERSTANDING

- All the feeders are pre decided with the allowed materials for feeding.
- As per the day plan and required materials, the bags are fed .
- All the materials feeding will be tracked in MES.
- The system shall be able to validate the feeding as per the correct feeder , FIFO and expired materials.
- Feeders will be identified with metal QR Code.

SOLUTION KEY FEATURES

- Feeders wise allowed materials and currently loaded materials will be displayed in HMI.
- MES Scanner will enable user to feed the materials while Andon light should indicate on wrong or right material scan .
- Andon lights will be integrated with PLC.

REQUIRED HARDWARE

- 2 Bluetooth wireless scanners required.
- 12 Andon lights.
- One HMI to display feeder wise material lookup.
- 12 Metal QRCode for each feeder.
- Network connectivity for all devices.
- Andon lights connectivity with existing or new PLC

OIL FEEDERS REQUIREMENT

UNDERSTANDING

- All the feeders are pre decided with the allowed oil for feeding.
- All the oil feeding will be tracked in MES.
- The system shall be able to validate the feeding as per the correct feeder , FIFO and expired materials.
- Feeders will be identified with metal QR Code.
- Oil feeding doors shall be controlled as per the feeding.

SOLUTION KEY FEATURES

- Feeders wise allowed oil and currently loaded oil will be displayed in HMI.
- MES Scanner will enable user to feed oil while Andon light should indicate on wrong or right oil scan.
- On wrong material scanned door should not open.
- Andon lights will be integrated with PLC.

REQUIRED HARDWARE

- 2 Bluetooth wireless scanners required.
- 10 Andon lights.
- One HMI to display feeder wise material lookup.
- 10 Metal barcode for each feeder.
- Network connectivity for all devices.
- Andon lights connectivity with existing PLC

AUTO CHEMICAL REQUIREMENT

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UNDERSTANDING

- WinCC SCADA is in place which is logging bag production and consumption.
- PLC or Scada has information about what production is planned , running and completed.
- Each trolly and feeder will be identified with unique QR Code.
- On wrong material scanned bin door should not open.

SOLUTION KEY FEATURES

- Feeders wise allowed material and currently loaded materials will be displayed in HMI.
- On wrong material scanned bin door should not open.
- One MES HMI display, to show trolly status , bin lookup and running and production history.
- Produced bags will be linked with input material based on consumption data received from SCADA.

REQUIRED HARDWARE

- 2 Wi-Fi mobile scanner for loading bins.
- 2 Bluetooth USB scanners required.
- 2 HMI to display .
- Network connectivity for all devices.

BALE CUTTER REQUIREMENT

UNDERSTANDING

- Bale cutters 2 different lots material cuts and can be mixed in one lot.
- After bale cutter final unique Lot, no will be generated and linked with source lot to trace the material.

SOLUTION KEY FEATURES

- MES will allow user to reprint the QRCode of cut bale for traceability.

REQUIRED HARDWARE

- 2 Printers.
- 2 USB scanners required.
- 2 HMI to display .
- Network connectivity for all devices.

MIXER FEEDING REQUIREMENT

REQUIREMENT

- Polymers and other materials in lot will be fed with scanning.
- Carbon and others material directly consumed from bins.
- Mixer control system PLC tags will have running production detail like plan id , recipe and material code, batch no etc.
- Since actual consumption is not in SCADA, MES will do consumption based on BOM quantity.

SOLUTION KEY FEATURES

- Mixer feeding area MES will monitor running material production plan and recipe.
- Based on the same BOM will be displayed to check material validation status.
- Before weighing user will scan polymer or compound.
- Conveyor will run in case of right material scanned and validated.
- All the input material will be linked with produced Batch ID.

REQUIRED HARDWARE

- 2 HMI display .
- 2 Wi- Fi Scanners.
- Network connectivity for all devices.

MIXER BATCH OFF REQUIREMENT

UNDERSTANDING

- The batch off at output will have defined PLC tag to inform when the batch off is completed with details.

SOLUTION KEY FEATURES

- Mixer Batch Off area MES will monitor batch off start and stop.
- On batch off stop user will enter no of batch taken out.
- MES is already tracking batches from mixed and dumped in FIFO from mixer production.
- MES will print the stickers for the batches taken out.

REQUIRED HARDWARE

- 2 HMI display .
- 2 printers
- Network connectivity for all devices.

SOLUTION ENGINEERING

Solution Proposal

- Overall system required MES for RMS to Mixer
  - To provide poka-yoke controls and alerts.
  - To provide traceability of Man, Material, Method and Machine.
  - To provide inventory information.
  - To integrate with existing systems.

- The system shall be planned in phase wise manner
  - Phase 1 : RMS to Mixer traceability (Only for Mixer materials)
  - Phase 2 : Mixer to FGS traceability (For all materials)

SCOPE OF WORK

PHASE 1 - MES

Master Data Management

1. Item management
2. BoM Management
3. User Management
4. Equipment management.

Work Order Execution

1. Tag printing
2. MHE mapping
3. Production & consumption

System Integrations

1. Oracle ERP
2. Scada , PLC

Quality Control

1. Error proofing & material validation
2. Material Aging and FIFO control

Warehouse

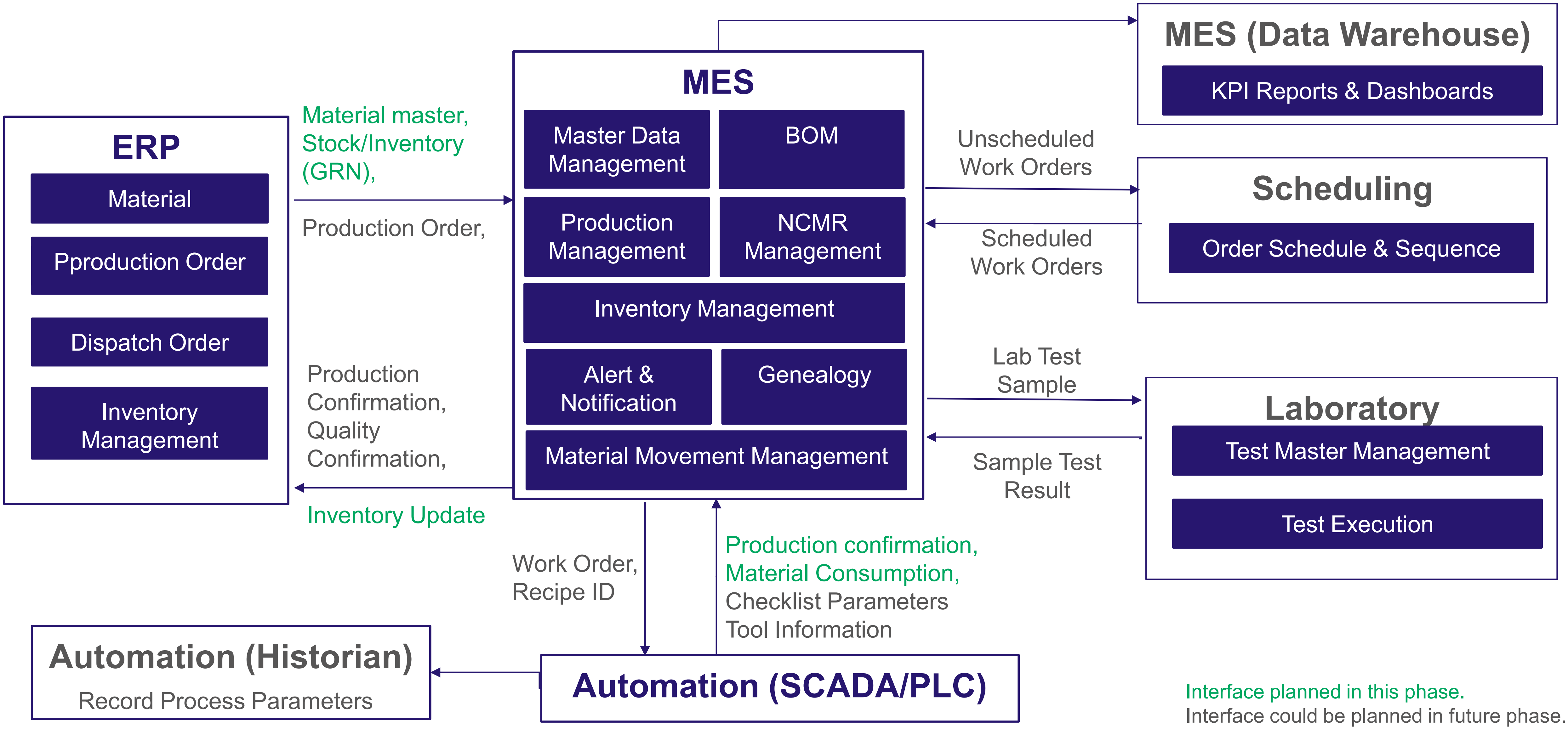
1. RM Booking
2. RM Issue

Reports

1. Production , consumption report
2. Traceability report

DATA FLOW ARCHITECTURE

MANUFACTURING OPERATION MANAGEMENT



PROPOSED SOLUTIONS

Ooverall system can be delivered in 2 different platform.

1

OpCenter Portfolio

- Siemens MOM multiple products catering to different module needs MES, AGW at one place.

2

Smart MES Portfolio

- Tailoring Smart MES solution to provide single solution catering all the requirement.

OPCENTER BASED

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OPERATION CENTRE PRODUCT PORTFOLIO

INDUSTRY SEGMENT

OP FN (OpCentre Framework)

- The architecture of OpCenter Execution Foundation exposing services to read or write data and execute commands while platform exposes all the required functionalities as services.

OP EX PR (OpCentre Execution Process)

- Formerly known as "SIMATIC IT Unified Architecture Process Industries" is Siemens' Manufacturing Execution System (MES) for Consumer-Packaged Goods, Food & Beverage, and Chemical industries.

OP EX DR (OpCentre Execution Discrete)

- Formerly known as "SIMATIC IT United Architecture Discrete Manufacturing") for complex assembly and job-shop environments.

OP EX MD (OpCentre Execution Medical Device)

- Formerly known as "Camstar Medical Device Suite" is in the medical device and diagnostics industry for error-proofing processes, paperless manufacturing, and electronic device history records (eDHR) and electronic batch records (eBR).

OP EX PH (OpCentre Execution Pharma)

- Formerly known as "SIMATIC IT eBR" is dedicated MES solution for the pharmaceutical industry

OP EX EL (OpCentre Execution Electronics)

- Formerly known as "Camstar Electronics Suite" is a full digital manufacturing solution for the electronics industry. It addresses PCB, mechanical assembly and box-build production.

OPERATION CENTRE PRODUCT PORTFOLIO

MOM MODULES

Opcenter Quality

- Formerly known as "QMS Professional" is a quality management system (QMS) that enables organizations to safeguard compliance, optimize quality, reduce defect and rework costs.

Opcenter APS

- Formerly known as "Preactor APS" Advanced Planning and Scheduling software solutions have been specifically developed to generate achievable production schedules.

OP RD&L (Opcenter Research, Development & Laboratory)

- Formerly known as "SIMATIC IT R&D Suite" for formula development, trial and experiment management.

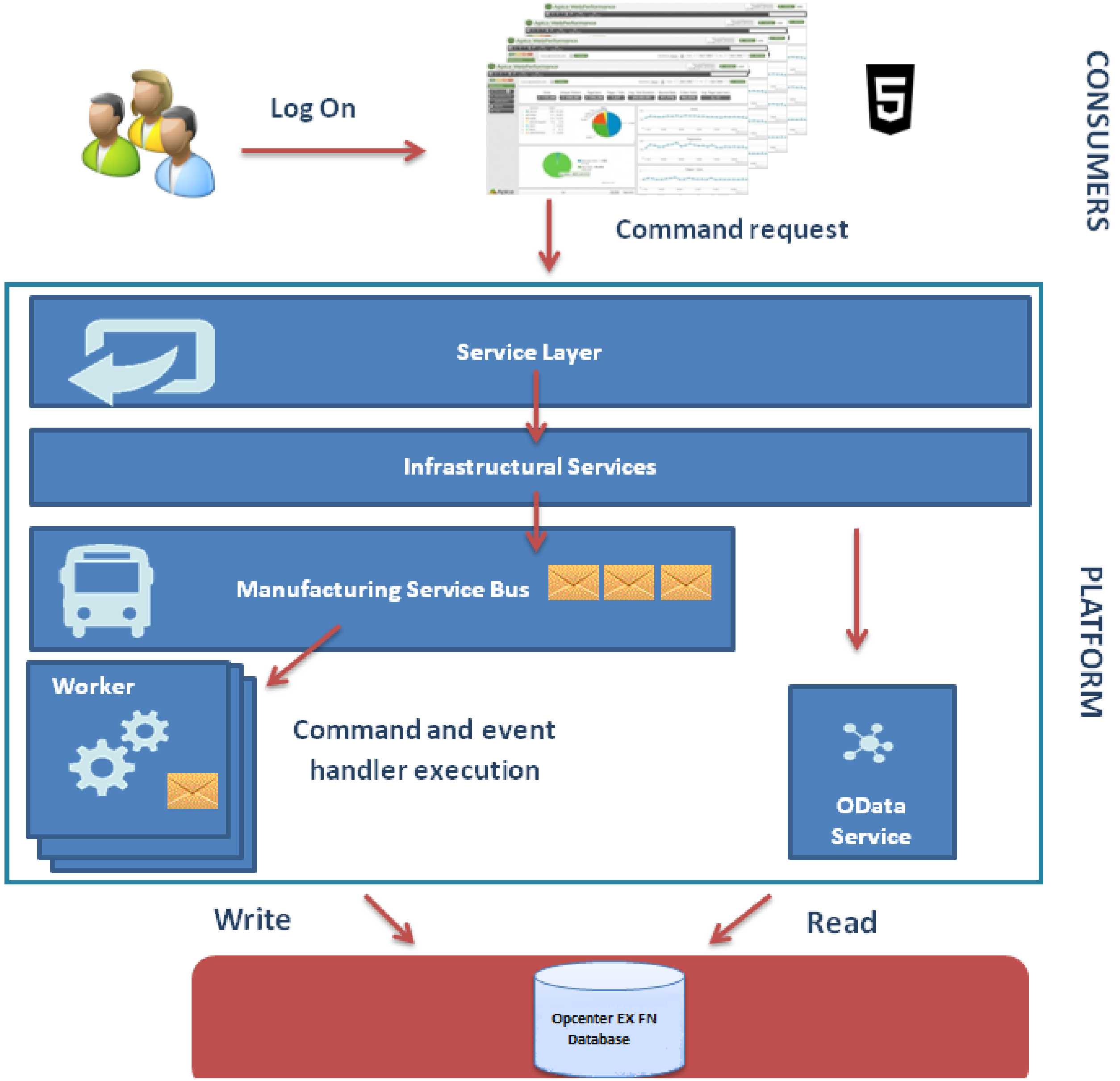
Opcenter Intelligence

- Formerly known as "Manufacturing Intelligence" is connects, organizes and aggregates manufacturing data from disparate company sources into cohesive, intelligent and contextualized information in order to gain immediate and actionable insights

OP CENTER FRAMEWORK ARCHITECTURE

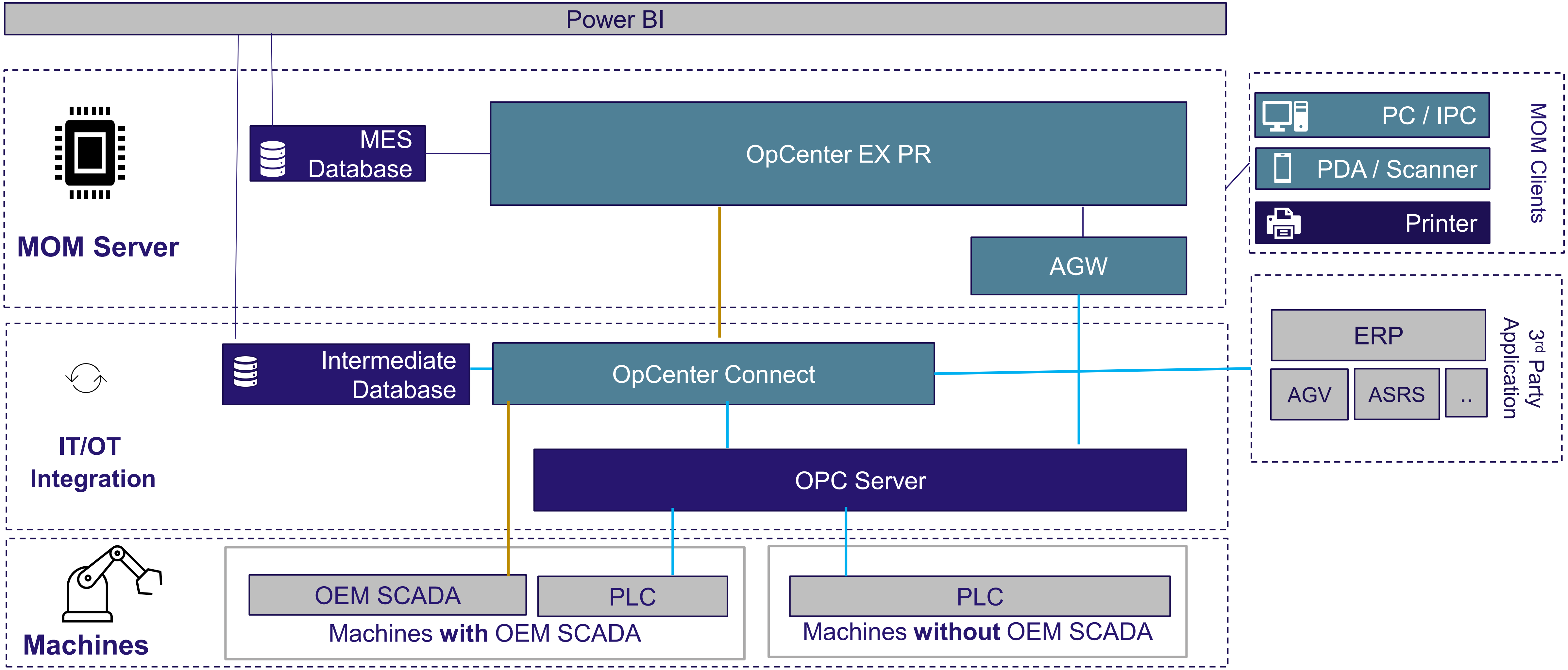
SERVICE ORIENTED ARCHITECTURE

- Service Layer
  - The command is received by the Service Layer, which queries the Manufacturing Services on whether a corresponding command exists and whether it is exposed through OData APIs.
- Infrastructural Services Layer
  - Identifies the worker role configured to process the command.
- Worker
  - An instance of the worker role executes the logic contained in the command handler.
- At the end of the transaction, an event is raised notifying all the operations that have been executed and the data that has been written on the database.



SOLUTION ARCHITECTURE

MANUFACTURING OPERATION MANAGEMENT



IMPLEMENTATION

COST COMPONENTS

Area	Component	Details	Owner	Cost Type
Software	Product License	EXPR, MIO , Kepware	Smart	Subscription
Software	3rd party software	OS, Antivirus, SQL, Visual Studio	Customer	Subscription
Hardware	Other components	Scanner, Printer, Clients	Smart	One Time
Hardware	Infrastructure	DEV,UAT and PROD	Customer	One Time
Service	Implementation & Support		Smart	One Time

Advantage

1. Globally accepted Siemens Products.
2. Multiple ready-to-use features.
3. Many other products in same eco system (MES, APS, Digital Twin, Teamcenter.)

Phase 1: Timeline

- Overall implementation : 4-5 months.
- Hyper care : 1 months.

SMART MES BASED

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PROPOSED SOLUTION - 2

Smart MES

- SMART MES is tailored solution consisting of Manufacturing Operation Management multiple modules.
- Providing one solution for complete Plant level plan , execution , control , monitoring and dashboards.
- Industrie 4.0 solution enables connecting with existing 3rd party systems, Level 4 ERP and Level 2 Machines for execution and reports generation which can be leveraged by end user for decision making.

Inclusive of MES, Smart MES Solution covers Light Weight features of,

1. WMS
2. LIMS – Integration
3. APS – Integration
4. CMMS
5. Others

INDUSTRY 4.0 TAILORED SOLUTION

Customer Complaint Management

NPD

Asset Management

PLM

WMS

Others

CRM

Fleet Management

Energy Utility Management

Tool Management

QBM

Smart MES Solution

Pressure on margins
- Need cost reduction
- Lesser Cost (Low Capex- Low Opex)

Extended Complexity
- Difficult to integrate systems & align business
- Simplified IT Landscape

Rapid changes in demand
- Economic turbulence (COVID)
- Regulation changes
- Integrated Solution easy to address changes

Greater eco-system
- across geographies, logistics
- Information availability

Optimize performance & revenues
- Across the entire value chain
- Informed decisions help in optimization

No EOL Reinstallation

Easy Maintenance

Easy Technology advancement. Cloud Ready

Faster rollout for onboarding another plant

TECHNOLOGY STACK

- React JS
  - Rich user interfaces
  - Fast rendering
  - Used by fortune 500
- .NET Core
  - Cloud friendly
  - High performance
  - High security
- React Native
  - High performance
  - Increased flexibility
  - Cross-platform
- SQL Server
  - Highly acceptable
  - Seamless compatibility
  - Cloud & Analytics friendly

Front End User Interface & Design

- React JS

Back End Business logic & processing

- .NET Core

Database

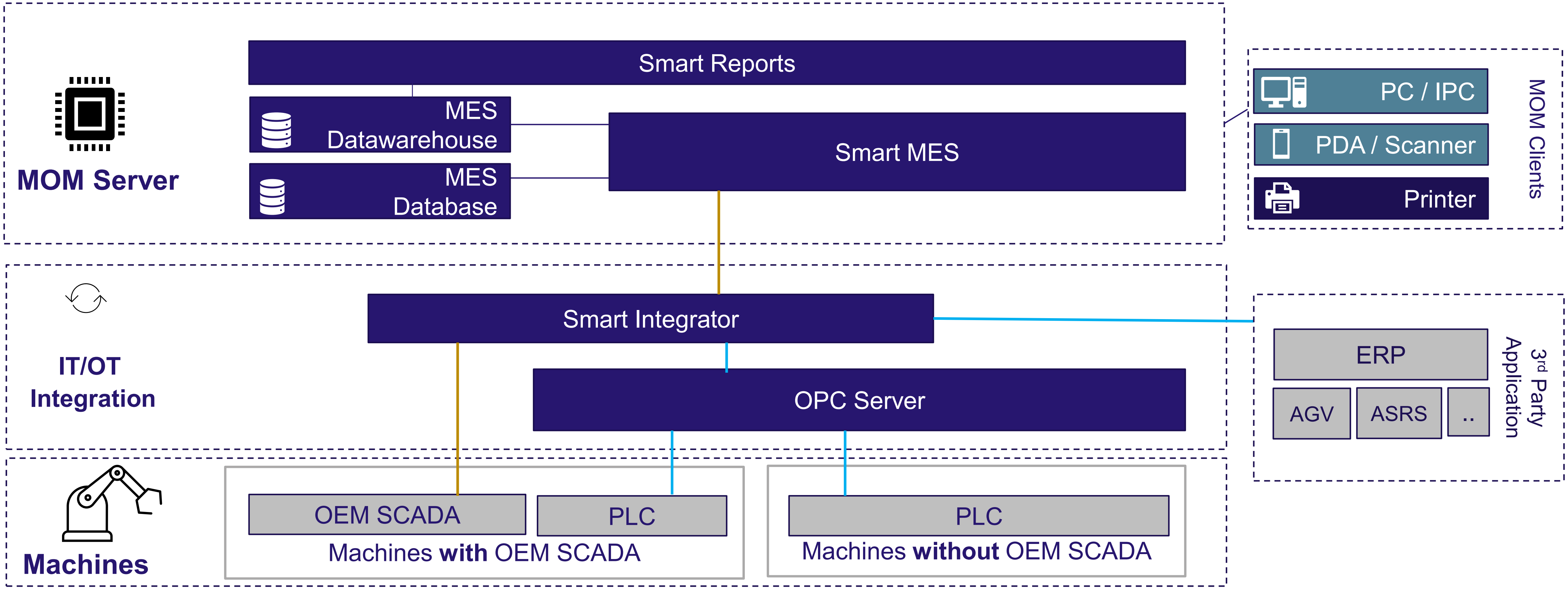
- MSSQL Server /PostgreSQL

Mobile Mobile application

- React Native

SOLUTION ARCHITECTURE

SMART MES



IMPLEMENTATION

COST COMPONENTS

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Area	Component	Details	Owner	Cost Type
Software	Product License	Smart MES	Smart	One Time
Software	3rd party software	OS, Antivirus, PostgreSQL,	Customer	One Time and Subscription
Hardware	Other components	Scanner, Printer, Clients	Smart	One Time
Hardware	Infrastructure	DEV,UAT and PROD	Customer	One Time
Service	Implementation & Support		Smart	One Time

Phase 1: Timeline

- Overall implementation : 4-5 months.
- Hyper care : 1 months.

Advantage

1. Single platform for all modules.
2 . Lightweight solutions with other module integrations.
3. Tailored exactly for the plant demand.

PROJECT EXCLUSIONS

1. Hardware & third-party Software supply is not included in current proposal
2. No IT infrastructure and networking analysis is included in the current proposal.
3. No legacy data will be migrated to MES during the project
4. Materials and activities on field devices and equipment (e.g., PLCs modification). Or modification of existing equipment for any process automation.
5. Providing and Coordinating warranty support for hardware, DB, OS, Hypervisor, SCADA, PLC or any third-party software / application.
6. No Hardware mounting & network configuration considered in current scope

PROJECT ASSUMPTIONS

1. Hardware Mounting and network will be in place.
2. Server IT infrastructure will be in place with all the devices connectivity.
3. ERP system is in place for raw material GRN ,Quality flow, reversal and inventory.
4. Based on material validation status , there will be provision in each machine control system that machine should not function in case of validation failed.
5. Some basic tags will be available in PLC like production start , stop , running recipe or material code plan id etc

IMPLEMENTATION BASIS

IMPLEMENTATION STAGES

MS No	Milestone Name	Key Activities	Deliverables
1	Requirement Freeze & Workshop	Plant visits & process understanding. Requirement gathering workshops.	BRD Signoff
2	Functional Design Session	Functional Design Documentation Interface Control Design Documentation	FDS Signoff ICD Signoff To-be UAC
3	Solution Build & SIT	Build & Developments Unit Testing & UAC testing SIT Deployment	UAC Signoff Solution Demo
4	User Acceptance Testing	UAT Deployment Solution test and acceptance test. Key User Training	UAT signoff
5	Commissioning & Go Live	Production Deployment	Go Live Acceptance Document
6	Roll Out	Handover training Training videos	Handover Training Training Videos

WAY AHEAD

NEXT STEPS

- ❑ Solution approach decision:
  - ❑ Smart MES / OpCenter MES
- ❑ Offer-
  - ❑ Technical Offer.
  - ❑ Techno Commercial Offer.

7 5 TOGETHER INTO THE FUTURE

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THANK YOU

Happy to resolve any queries @

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